

Complaint workload: a two-timescale reporting reference

A compact blend of recent and slower reporting histories improves reserved-month complaint-count prediction over matched simple and flexible controls. It predicts administrative workload within known vehicle groups, not defect risk or mechanical reliability.

Data and evaluation

NHTSA 2025-2026 public vehicle complaints are deduplicated by ODINO and make/model across component records. The 434 vehicle cohorts have at least six received complaints in Jan-Jun 2025, fixed before all target months. Development Jul 2025-Apr 2026 uses 4340 cohort-monththrows; confirmation May-Jul 2026 uses 1302 rows in three complete months. Zero counts retained.

Target: **Next-month unique received vehicle complaints per known make/model**, measured as complaints/month. Primary error: complaints/month. Only prior-month LDATE received counts predict next month. Retrospective snapshot may revise product labels and records; historical DATEA/report publication lag means calendar causality is not verified as-issued availability. No incident-date future information used.

Whole target months; same known vehicle cohorts repeat, lag windows overlap. Three held months are not independent fleets or defect experiments. Cohort selected before target period using ≥ 6 received Jan-Jun 2025 complaints. Development targets Jul 2025-Apr 2026, rolling forward blocks; May-Jul 2026 confirmation. No target-month counts enter predictors.

Findings and equations

$$\hat{N} = \text{clip}(L_3 + a(L_1 - L_3) + b(L_3 - L_6), 0, 10^8)$$

The notation and all fitted coefficients are defined below.

P100-019-F01 - validated_extension (partially_supported). A compact two-timescale reporting-history blend improves mean reserved-month count error over the frozen matched controls.

$\text{count_hat} = L_3 + a(L_1 - L_3) + b(L_3 - L_6)$, with nonnegative clipping.

Recent observations and a slow historical level capture persistence with transient reporting variation. The final weights are approximately convex, but their mechanism and cross-period stability are not identified.

Falsifying evidence and limits: Only three held months, known cohorts and retrospectively revised labels. Other unselected states can better this particular confirmation; no industrial cost benefit demonstrated.

P100-019-F02 - informative_falsification (supported). The campaign does not identify a unique nonlinear reporting-burst law or a vehicle defect-rate relationship.

Asymmetric 003, variance-stabilized 004/005 and robust 006 remain traceable.

Square-root stabilization modestly improves development mean but does not displace simpler 002 underpredeclared 1 percent complexity rule.

Falsifying evidence and limits: No population vehicle exposure, verified failure target or causal recall intervention. A complaint count is not failure probability.

Selected executable model: `attempt_002`. Let L_1 be last month count and L_3 , L_6 trailing three/six-monthly means. Selected $\text{count_hat} = \text{clip}[L_3 + a(L_1 - L_3) + b(L_3 - L_6), 0, 1e8]$, $a = 0.5971048228619386$, $b = -0.36844397185306554$. The final fitted form equals approximately $0.5971L_1 + 0.03445L_3 + 0.36844L_6$. These are fitted descriptive memory weights, not a universal reporting or failure-rate kernel.

All coefficients, input definitions, training groups and transformations are in `rules.json`; the runnable reference performs no fitting.

Compact numerical evidence

Errors are computed within declared complete groups (with final survey weights when supplied), then averaged equally across groups.

Frozen model	Development error	Confirmation error	Worst confirmation group
reference	3.955552	3.511835	3.742832
constant	15.9496	15.64825	15.75912
flexible	6.647887	3.77217	3.943177
persistence	4.302995	3.652842	3.868664
trailing 3	4.159498	3.769585	4.012289
trailing 6	4.398105	4.075269	4.324117

Confirmation contains 1302 scored observations in 3 groups. Individual-group scores and every attempted model remain available. Confirmation MAE 3.511835 complaints pervehicle/month versus persistence 3.652842, trailing 3=3.769585, trailing 6=4.075269, flexible 3.772170 and constant 15.648246. Worst held month 3.742832. Several more complex unselected attempts score better retrospectively; they remain unselected. Three adjacent months do not justify narrow population confidence intervals.

Interpretation, limitations and use

Administrative complaint workload among known product VMAKE/model labels, not vehicle failures or safety risk. No fleet-at-risk denominator, reporting propensity or causal recall effect; deduplicated ODINO/vehicle across components, snapshot relabeling remains. Whole-group errors/ranges; no row bootstrap or manufactured independent-population confidence. Confirmation becomes exposed after freeze. Useful for comparing simple interpretable administrative workload forecasts under whole-month validation. The data have no vehicle fleet-at-risk denominator or measured defect incidence. Reporting propensity and revised product labels prevent causal recall or reliability conclusions; actual staffing/industrial savings were not tested.

Reproducibility and prior art

Run python run.py to verify hashes and replay every frozen equation. Predictions use declared input columns only. The shared benchmark evaluator can score alternative equations under the same task; numerical agreement with these coefficients is not required. Public source-aware confirmation is now exposed. No independent experimental replication or certified novelty is claimed.

- Complaints identify potential issues together with other sources, not verified incidence. Native CMPL documentation defines received date and duplicate ODINO components; autoregressive reporting processes are existing baseline families. (checked 2026-10-02).